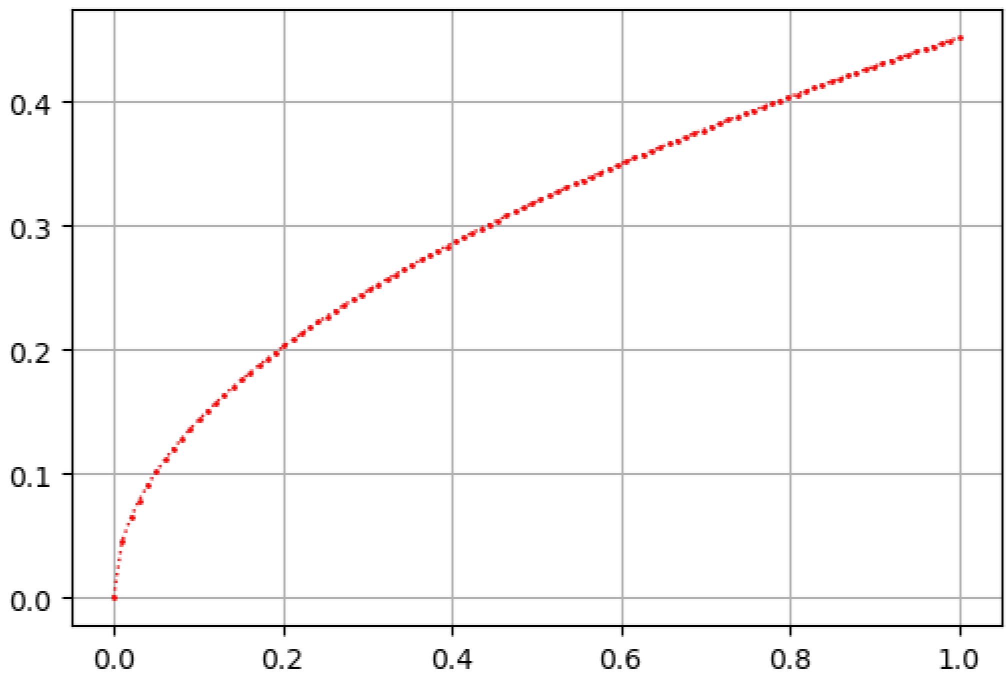
In [6]:

```
import tkinter as tk
from tkinter import ttk
import matplotlib.pyplot as plt
from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
import numpy as np

def plot_logistic_map():
    x = np.linspace(0,1,100)
    g = 9.8
    y = np.sqrt(2*x/g)
    plt.plot(x,y,'o',linestyle = 'dotted',color = 'r',linewidth = '1',markersize =
    plt.grid()
    canvas.draw()

def clear_plot():
    fig.clf()
    canvas.draw()

# Main Window
root = tk.Tk()
root.title("Logistic Map Visualizer")
root.geometry("700x600")
root.configure(padx=10, pady=10)

# Input Fields
ttk.Label(root, text="Enter x min value:").grid(row=0, column=0, sticky='w')
entry_r = ttk.Entry(root)
entry_r.insert(0, "3.5")
entry_r.grid(row=0, column=1, sticky='ew')

ttk.Label(root, text="Enter x max value:").grid(row=1, column=0, sticky='w')
entry_x0 = ttk.Entry(root)
entry_x0.insert(0, "0.1")
entry_x0.grid(row=1, column=1, sticky='ew')

# Buttons
plot_button = ttk.Button(root, text="Plot", command=plot_logistic_map)
plot_button.grid(row=2, column=0, pady=5)

clear_button = ttk.Button(root, text="Clear", command=clear_plot)
clear_button.grid(row=2, column=1, pady=5)

# Plotting Area
fig = plt.figure(figsize=(6, 4))
canvas = FigureCanvasTkAgg(fig, master=root)
canvas.get_tk_widget().grid(row=3, column=0, columnspan=2, pady=10)

root.mainloop()
```



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